



Track: AI & Data Science



DriveGuard

AI In-Cabin Driver Monitoring System



AI powered real-time in-cabin monitoring that authenticates drivers, detects distractions, drowsiness, and threats, and logs all events through a unified dashboard.

01



Problem

- Lack of integrated driver safety systems
- Increased risk of road accidents
- Distraction, drowsiness, unauthorized use, and delayed response

02



Our Solution

- Real-time AI in-cabin monitoring
- Multi-threat detection in one system
- Unified dashboard & event logging

03



Core AI Modules

- **Authentication** (Driver Verification)
- **Distraction Detection** (Phone, Cigarette, Vape, Bottle)
- **Drowsiness Detection** (Sleepiness & Fatigue)
- **Violence Detection** (Weapons & Physical Assault)

04



Key Outputs

- ✓ Secure driver authentication with anti-spoofing
- ✓ Real-time multi-threat detection
- ✓ Instant multi-level alerts
- ✓ Centralized dashboard & monitoring
- ✓ Automated event logging

05



Stakeholders

- Drivers
- Transport Companies
- Insurance Companies
- Car Manufacturers

06

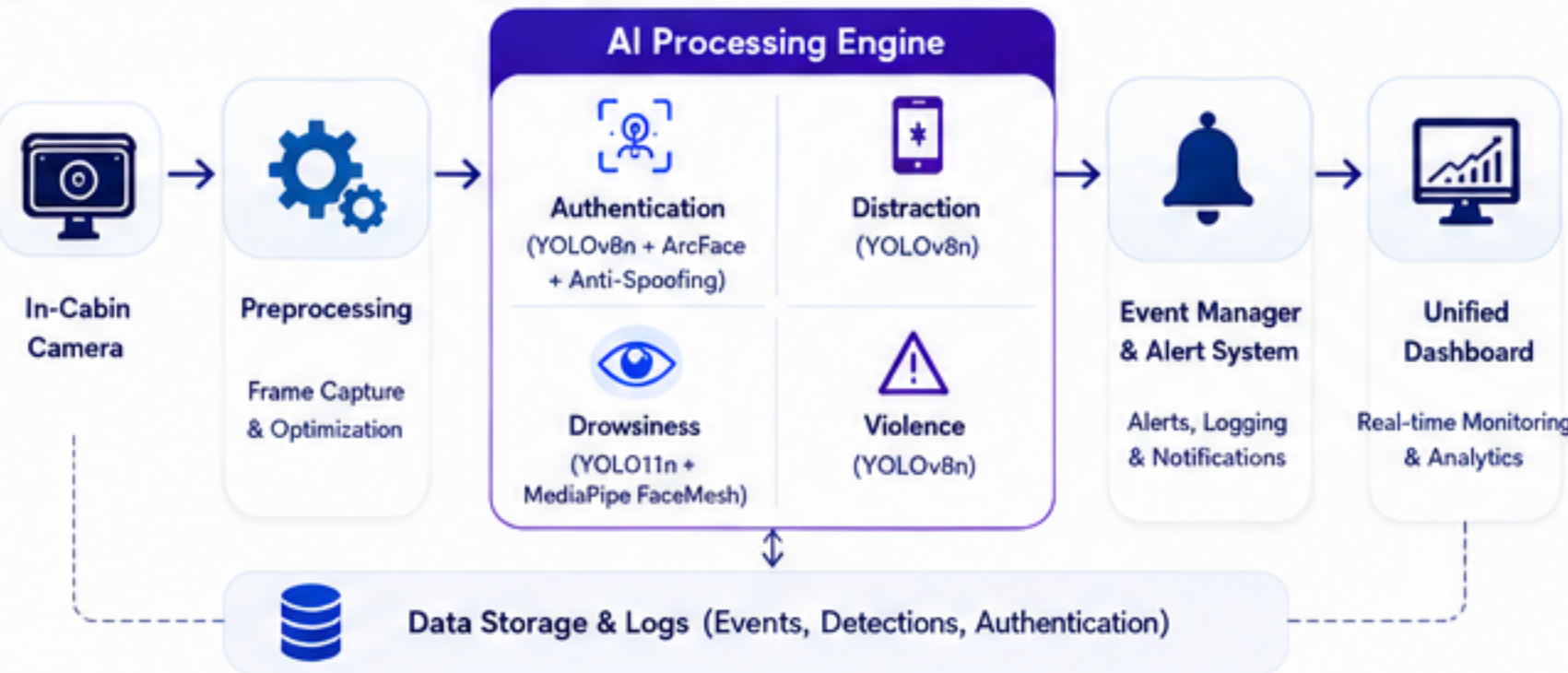


Impact

- ✓ Safer roads
- ✓ Reduced accidents
- ✓ Accountability & compliance
- ✓ Operational efficiency



SYSTEM ARCHITECTURE



KPIs (Performance)

	Authentication AUC $\approx$ 0.99	
	Distraction Detection mAP50 = 0.77	Bottle: 0.941 Phone: 0.889 Cigarette: 0.700 Vape: 0.550
	Drowsiness Detection mAP50 = 98.4%	Precision: 95.9% Recall: 95.0% ~2.8 ms/frame
	Violence Detection F1 = 82.8%	Weapon mAP50: TBD



TECHNOLOGY STACK

Languages & Frameworks		
Python 3.11	Flask	HTML5
AI / ML Models & Libraries		
YOLOv8/11	ArcFace	MediaPipe
APIs & Streaming		
Flask REST API	Flask-CORS	MJPEG Live Streaming
Tools & Infrastructure		
SQLite	MLflow	NVIDIA GTX 1650 (CUDA)



TEAM

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DEPLOYMENT

Python 3.11 YOLOv8/11 ArcFace, MediaPipe, MoViNet	CAP Flask REST API (Backend Services)	HTML Dashboard (Real-time Monitoring)
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PROJECT OUTCOMES

Real-time Monitoring System	AI Models for Multi-threat Detection	Unified Dashboard	REST API Services	Automated Event Logging
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DriveGuard enhances driver safety through AI-powered authentication, real-time monitoring, and intelligent threat detection — for safer roads and smarter mobility.